

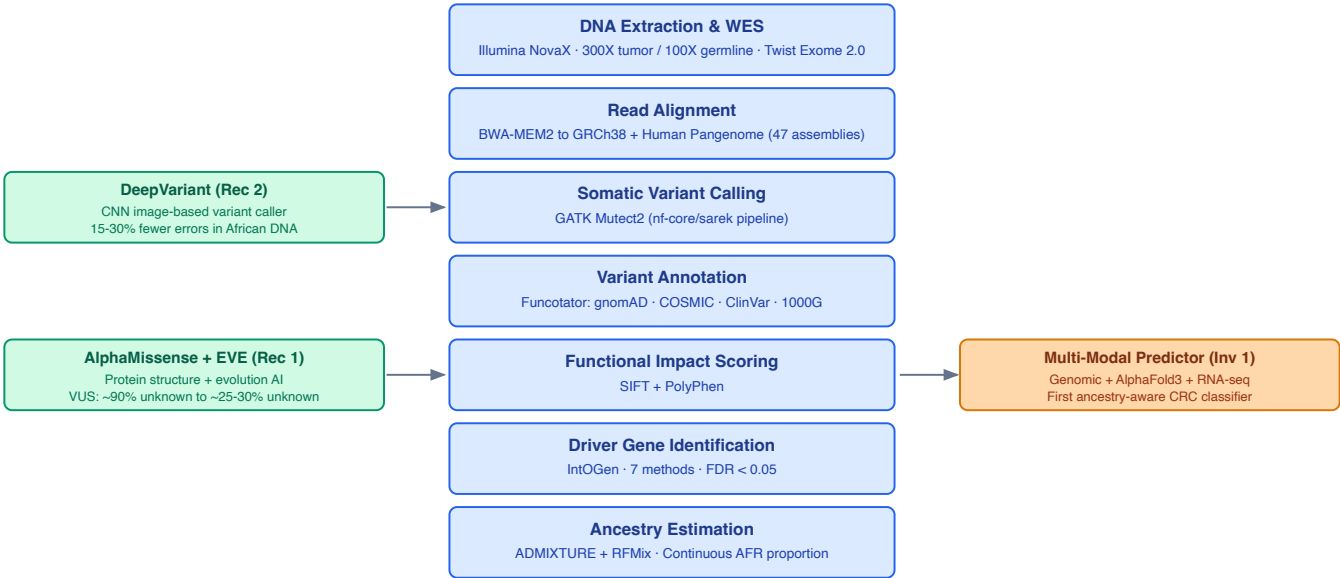
AI Integration at Each Point of the AfriCAN DANCE Scope of Work

Mapped to the R01 Domestic Project Research Strategy (PF5) · July 2026

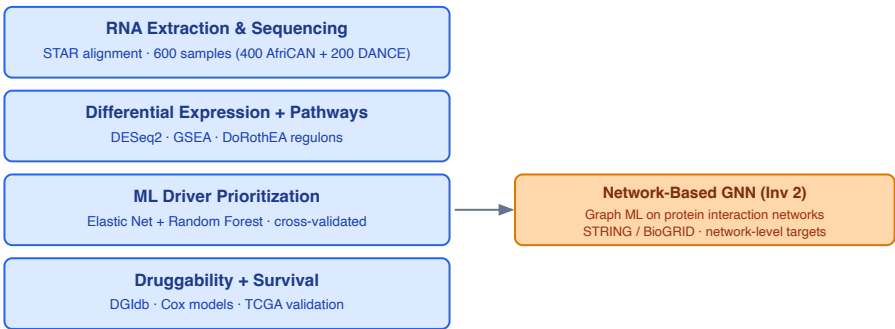
AI Integration Points in the AfriCAN DANCE Study Pipeline

Blue = Grant pipeline step · Green = AI recommendation (apply existing tools) · Orange = AI innovation (build new system)

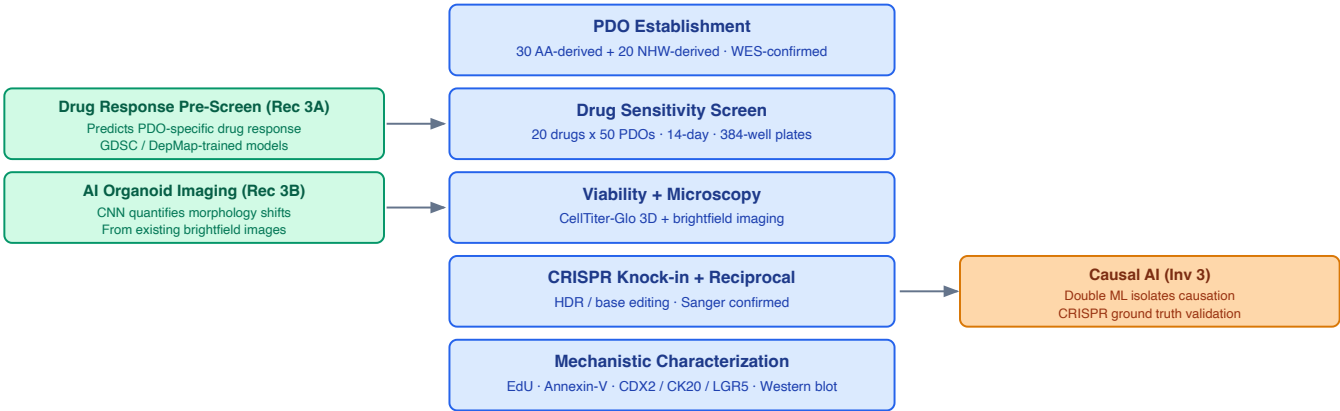
AIM 1: DISCOVERY — Whole-Exome Sequencing (n = 1,400 tumor-normal pairs)



AIM 2a: TARGETS — RNA-seq + Machine Learning (n = 600 transcriptomes)



AIM 2b: VALIDATION — Patient-Derived Organoids + CRISPR



Grant pipeline step AI recommendation AI innovation

Recommendation 1: AlphaMissense + EVE — Variant Reclassification

Aim 1 · Functional Impact Scoring

Grant location: Aim 1 Methods, "Somatic variant calling and annotation" section. The grant states: *"Functional impact scores will be estimated using SIFT and PolyPhen."*

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Integration: Run AlphaMissense (Google DeepMind protein-structure predictor) and EVE (evolutionary conservation model) alongside SIFT and PolyPhen as a multi-tool consensus. AlphaMissense has pre-scored all 71 million possible missense variants. EVE provides an orthogonal evolutionary signal. Neither requires the variant to appear in any clinical database, directly addressing the circular problem of European-ancestry-trained classifiers.

Expected impact: Reclassifies approximately 60-70% of the 87-92% of variants currently classified as "unknown" in African-ancestry tumors. Feeds reclassified variants downstream into the IntOGen driver gene identification pipeline, improving both sensitivity and specificity of driver detection. Timeline: 2-3 weeks, one bioinformatician.

Recommendation 2: DeepVariant — AI Variant Calling

Aim 1 · Somatic Variant Calling

Grant location: Aim 1 Methods, "Somatic variant calling and annotation" section.

"Somatic variant calling on tumor-normal matched pairs will be performed using GATK Mutect2 within the nf-core/sarek pipeline."

Integration: Run DeepVariant (Google CNN-based caller) in parallel with Mutect2 on both GRCh38 and pangenome alignments. The grant already addresses reference bias by mapping to the human pangenome (47 diploid assemblies). DeepVariant complements this at the caller level: it converts read pileups into image tensors and uses a neural network trained on millions of examples, reducing false positives and false negatives by 15-30% in non-European samples. Cross-reference Discordant calls between Mutect2 and DeepVariant as high-confidence variant set.

Expected impact: 15-30% reduction in false variant calls in African-ancestry DNA. Particularly improves indel detection in repetitive and structurally variable regions where Mutect2 struggles. Timeline: 1-2 weeks.

Recommendation 3A: Drug Response Pre-Screen — Computational Prioritization

Aim 2b · Drug Sensitivity Screen (pre-screen phase)

Grant location: Aim 2b Methods, "Variant-stratified primary drug sensitivity screen" section.

"All 50 PDO lines will be treated with the top 20 candidate inhibitors nominated by Aim 2a bioinformatic pipeline, plus vehicle control and current standard-of-care regimens (FOLFOX), across a range of concentrations."

Integration: The 20 drugs are already computationally nominated by the Aim 2a pipeline (Elastic Net + Random Forest). The AI pre-screen adds a distinct layer: predicting *organoid-specific* drug response before wet-lab work begins. Models trained on GDSC, CCLE, and DepMap pharmacogenomic databases can forecast which of the 50 PDO lines are most likely to show ancestry-specific differential response to each drug. This allows the team to prioritize the highest-value PDO-drug combinations,

optimize screening order, and flag combinations likely to yield the strongest ancestry-stratified signals before committing 14-day experiments.

Expected impact: Prioritizes wet-lab resources toward highest-value drug-PDO combinations. Does not replace the full screen but focuses attention where ancestry-specific effects are most likely. Timeline: 2-3 months to build and validate the model.

Recommendation 3B: AI Organoid Imaging — Morphological Analysis

Aim 2b · Viability + Microscopy Assessment

Grant location: Aim 2b Methods, "Variant-stratified primary drug sensitivity screen" section.

"Treatment will continue for 14 days or until significant differences in organoid viability, growth rate, or morphology are observed, assessed by microscopy and CellTiter-Glo 3D Cell Viability Assay."

Integration: Brightfield microscopy is already part of the grant protocol. However, without AI analysis, these images serve only qualitative or semi-quantitative purposes. A convolutional neural network (CellPose for segmentation, followed by CNN feature extraction) converts existing brightfield images into quantitative morphological features: organoid size distribution, boundary irregularity, internal structure complexity, and invasion patterns. This captures drug effects that alter organoid architecture without killing the cell, which CellTiter-Glo viability assays cannot detect.

Expected impact: Transforms existing microscopy data from qualitative observation into multi-dimensional quantitative drug response profiles. Captures structural drug effects invisible to viability-only readouts. Timeline: concurrent with drug screen, plus 2-3 months for analysis pipeline.

Innovation 1: Custom Multi-Modal Variant Effect Predictor

Spans Aim 1 (WES) + Aim 2a (RNA-seq) + Aim 2b (organoid data)

Scope overlap: Extends beyond a single pipeline step. Combines data from three sources that no existing tool integrates:

- **Genomic context** from Aim 1 WES: variant position, type, conservation score, trinucleotide context
- **Structural context** from AlphaFold3: 3D protein location, domain, interface proximity, solvent accessibility
- **Transcriptomic context** from Aim 2a RNA-seq: expression impact, pathway dysregulation, regulon activity

Training labels come from ClinVar/COSMIC known variants (transfer learning) plus the study's own Aim 2b organoid drug response data (gold-standard functional validation). This is the only dataset in the world with matched somatic mutations, RNA-seq, clinical outcomes, and organoid drug response from African-ancestry CRC tumors.

Expected impact: First ancestry-aware, CRC-specific AI variant classifier. Publishable as a standalone methods paper. If it does not outperform AlphaMissense/EVE, the Tier 1 results still stand (de-risked). Timeline: 4-6 months, dedicated ML engineer.

Innovation 2: Network-Based AI for Drug Target Discovery

Aim 2a · ML Driver Prioritization

Grant location: Aim 2a Methods, "Prioritization of druggable oncogenic drivers" section.

"We will employ two machine learning algorithms... elastic net regression, as well as random forest modeling. These methods

function through distinct processes with complementary sets of advantages and limitations."

Integration: Replaces or extends single-gene Elastic Net + Random Forest analysis with a Graph Neural Network (GNN) operating on protein-protein interaction networks (STRING, BioGRID). Genes do not work in isolation. A gene that looks unremarkable in isolation may be a critical hub in a network neighborhood collectively rewired in African-ancestry tumors. The Elastic Net and Random Forest evaluate each gene independently. The GNN evaluates entire network neighborhoods, identifying collective dysregulation patterns that single-gene methods miss entirely.

Expected impact: Identifies drug targets by network neighborhood rather than individual gene ranking. Captures network-level therapeutic vulnerabilities invisible to standard approaches. Timeline: 3-4 months, graph ML expertise.

Innovation 3: Causal AI for Treatment Response

Aim 2b · CRISPR Validation analysis

Grant location: Aim 2b Methods, "Variant-stratified primary drug sensitivity screen" analysis and "CRISPR knock-in" validation sections.

"The primary analysis for each candidate inhibitor will compare drug sensitivity between PDO lines carrying the matched ancestry-enriched driver variant and lines that do not, using a mixed effects model with variant carrier status as the primary predictor."

Integration: The grant uses mixed effects models to associate variant carrier status with drug response. Causal AI (Double Machine Learning) goes further: it isolates whether a specific mutation *causes* differential drug response or is merely correlated with ancestry-linked features. Example: if KRAS G12D (more common in AA tumors) is associated with worse drug response, is that because KRAS G12D itself causes resistance, or because it co-occurs with other ancestry-linked factors? The CRISPR knock-in and reciprocal reversion experiments in Aim 2b provide exactly the experimental ground truth needed to validate these causal predictions, creating a tight computational-to-experimental loop.

Expected impact: Strengthens causal claims from observational associations. CRISPR experiments provide ground truth validation. Significantly tightens the evidence chain from variant to drug response. Timeline: 3-5 months, causal inference expertise.